

Supplementary Technical Report: Proxy-Guided Sampling for Approximate Graph Aggregation with Machine Learning Predicates

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This technical report serves as a supplementary document to our main submission. Due to strict space limitations in the conference format, we have deferred the detailed real-world business case study (Section 1) and the rigorous mathematical proof of uniform tree-homomorphism sampling (Section 2) to this document.

1 Real-World Motivation and Business Case Study

This section details a concrete, high-value business scenario demonstrating that Graph Aggregation with ML predicates (GA-ML) is an urgent requirement in modern AI-augmented databases, rather than an artificial extension of standard graph queries.

The Real-World Need for Multi-Modal ML on Graphs. Modern enterprises (e.g., Amazon, Alibaba) manage their data as heterogeneous property graphs (comprising User, Product, and Review vertices). However, crucial business insights are often locked inside unstructured attributes (images and text) attached to these vertices. For instance, a quality-control analyst investigating user dissatisfaction needs to query: *"Find all transactions where a user purchased a wooden furniture product and left an angry/negative review."*

Traditional relational filters cannot express this. It strictly requires multi-modal ML predicates: a Computer Vision (CV) model to verify "wooden furniture" from product images, and an NLP sentiment model to detect "angry" emotions in review texts.

The Business Value of Graph Aggregation (COUNT/SUM/AVG). In enterprise scenarios, merely retrieving a massive list of matched subgraphs is unhelpful for decision-making. Analysts fundamentally need macro-level aggregations over these topological structures. As detailed in our revised Introduction, applying an aggregation function λ provides distinct business KPIs:

- $\lambda = \text{COUNT}$: Quantifies the total transaction volume of these controversial orders.
- $\lambda = \text{SUM}$: Calculates the total sales revenue at risk (by aggregating the price attribute of the matched subgraphs).
- $\lambda = \text{AVG}$: Identifies the average unit price of the criticized items, helping to pinpoint if the issue lies in high-end or low-end markets.

2 Proof of Uniform Tree-Homomorphism Sampling

In Section IV-C of the main paper, we proposed a top-down sampling procedure to draw a tree homomorphism $h \in \Omega$ over the Candidate Space (CS) summary. Below, we formally prove that this procedure ensures every valid homomorphism in Ω is sampled uniformly at random with exact probability $1/|\Omega|$.

Sampling Procedure Recap. Let T be a spanning tree of the query graph Q with root r . The number of homomorphisms of the subtree of T rooted at u , given u is mapped to v , is recursively calculated via dynamic programming:

$$H(u, v) = \begin{cases} 1 & \text{if } u \text{ is a leaf in } T \\ \prod_{u' \in \text{ch}(u)} \sum_{v' \in C(u' | u \rightarrow v)} H(u', v') & \text{if } u \text{ is an internal node} \end{cases} \quad (1)$$

The total number of homomorphisms is $|\Omega| = \sum_{v \in C(r)} H(r, v)$.

Our sampling procedure draws $h(r) = v_r \in C(r)$ with probability $P(r \rightarrow v_r) = \frac{H(r, v_r)}{|\Omega|}$. Then, iteratively for each mapped query vertex $u \rightarrow v$ and each of its unmapped children $u' \in \text{ch}(u)$, it selects a candidate data vertex $v' \in C(u' | u \rightarrow v)$ with probability proportional to $H(u', v')$:

$$P(u' \rightarrow v' | u \rightarrow v) = \frac{H(u', v')}{\sum_{v'' \in C(u' | u \rightarrow v)} H(u', v'')} \quad (2)$$

Theorem 1 (Uniform Tree Sampling). *For any complete valid tree homomorphism $h \in \Omega$, the probability of generating h using the proposed top-down sampling procedure is exactly $P(h) = \frac{1}{|\Omega|}$.*

Proof. Let T_u denote the subtree rooted at node u . We first prove by structural induction (bottom-up) that the conditional probability of sampling the exact partial homomorphism $h|_{T_u}$ for the subtree T_u , given that its root u is already mapped to $h(u)$, is exactly $\frac{1}{H(u, h(u))}$. Let this conditional probability be denoted as $P(T_u \text{ matches } h | u \rightarrow h(u))$.

Base Case: If u is a leaf in T , there are no children to map. The subtree T_u consists only of u . Given $u \rightarrow h(u)$, the mapping is trivially complete with probability 1. By definition, $H(u, h(u)) = 1$. Thus, $P(T_u \text{ matches } h | u \rightarrow h(u)) = 1 = \frac{1}{1} = \frac{1}{H(u, h(u))}$. The base case holds.

Inductive Step: Assume u is an internal node, and the property holds for all of its children $u' \in \text{ch}(u)$. The probability of generating the exact subtree mapping $h|_{T_u}$ given $u \rightarrow h(u)$ is the product of the probabilities of independently mapping each child $u' \rightarrow h(u')$ and recursively matching their respective subtrees $T_{u'}$.

$$P(T_u \text{ matches } h | u \rightarrow h(u)) = \prod_{u' \in \text{ch}(u)} \left[P(u' \rightarrow h(u') | u \rightarrow h(u)) \times P(T_{u'} \text{ matches } h | u' \rightarrow h(u')) \right]$$

By substituting the sampling probability $P(u' \rightarrow h(u') | u \rightarrow h(u))$ and applying the inductive hypothesis $P(T_{u'} \text{ matches } h | u' \rightarrow h(u')) = \frac{1}{H(u', h(u'))}$, we obtain:

$$\begin{aligned} P(T_u \text{ matches } h | u \rightarrow h(u)) &= \prod_{u' \in \text{ch}(u)} \left[\frac{H(u', h(u'))}{\sum_{v'' \in C(u' | u \rightarrow h(u))} H(u', v'')} \times \frac{1}{H(u', h(u'))} \right] \\ &= \prod_{u' \in \text{ch}(u)} \frac{1}{\sum_{v'' \in C(u' | u \rightarrow h(u))} H(u', v'')} \\ &= \frac{1}{\prod_{u' \in \text{ch}(u)} \sum_{v'' \in C(u' | u \rightarrow h(u))} H(u', v'')} \end{aligned}$$

By the recursive definition of $H(u, v)$ from Eq. 1, the denominator is precisely $H(u, h(u))$. Therefore:

$$P(T_u \text{ matches } h \mid u \rightarrow h(u)) = \frac{1}{H(u, h(u))}$$

This completes the induction.

Final Probability Calculation: The entire homomorphism h is formed by first mapping the root r to $h(r)$ and then recursively mapping the subtree T_r . The total probability $P(h)$ is:

$$\begin{aligned} P(h) &= P(r \rightarrow h(r)) \times P(T_r \text{ matches } h \mid r \rightarrow h(r)) \\ &= \frac{H(r, h(r))}{|\Omega|} \times \frac{1}{H(r, h(r))} \\ &= \frac{1}{|\Omega|} \end{aligned}$$

Thus, every valid homomorphism $h \in \Omega$ is sampled with equal probability $\frac{1}{|\Omega|}$. This rigorous bottom-up structural induction completes the proof of uniform sampling. $\square$